Kent Pearce

Foundations of Algebra I

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- Syllabus: *3360-20101(001)-syl.pdf*
- Schedule: *3360-20101(001)-cal.pdf*
- Homework: *3360-20101(001)-hwk.pdf*

Office Hours: MW 2:00 - 4:00 or
By Appointment

MATH 3360-001

Text
Papantonopoulou, A.

Algebra, Pure & Applied
Prentice-Hall

Lecture
Room: MA 012
Time: MWF 11:00-11:50

Learning Objectives

Learning Outcomes: Students learn how to think and reason abstractly in the context of algebraic structures, and learn how to write correct and clear mathematical arguments in this context. Concepts to be mastered by the students include but are not limited to the following:

a. Groups and group homomorphisms
b. Group actions
c. Rings and ring homomorphisms
d. Polynomials

Assessment of the learning outcomes

Assessment will be achieved through one or more activities, non-graded and graded, such as: class attendance, class discussion, board work, short quizzes, selected homework, writing assignments, examinations and other optional activities deemed appropriate by the instructor. Class grades will be assigned according to the following rubric:

Curricular Content

Chapter 0 – Background

a. Sets, Maps & Functions
   Concepts: One-to-one, Onto, Image, Invertible
b. Equivalence Relations & Partitions
   Concepts: Reflexive, Symmetric, Transitive
c. Properties of Integers
   Principles: Well-Ordering Axiom, Mathematical Induction, Binomial Theorem, Division Algorithm, Euclidean Algorithm, Fundamental Theorem of Arithmetic, Modular Arithmetic
   Concepts: Greatest Common Divisor, Least Common Multiple, Linear Combination

d. Complex Numbers
   Concepts: Real Part, Imaginary Part, Modulus, Conjugate,
e. Matrices
   Concepts: Matrix, Matrix Sum, Matrix Product, Invertible, Determinant

Chapter 1 – Groups

a. Examples and Basic Concepts
   Concepts: Operation, Group Axioms, Abelian, Group Tables, Order
   Principles: Basic Group Properties
b. Subgroups
   Concepts: Subgroup, Cyclic Subgroup, Center
c. Cyclic Groups
   Concepts: Cyclic, Generator, Finite Order, Subgroup Lattice
d. Permutations
   Concepts: Permutation, Permutation Group (Symmetric Group of Degree *n*) S_n, Length, Cycle, Alternating Group of Degree *n*

Chapter 2 – Homomorphisms

a. Cosets and Lagrange's Theorem
   Concepts: Cosets, Index
   Principles: Lagrange's Theorem
b. Homomorphisms
   Concepts: Homomorphim, Isomorphism
   Principles: Basic Group Homomorphism Properties
c. Normal Subgroups
   Concepts: Normal Subgroup
d. Quotient Groups
   Concepts: Quotient Group
   Principles: First Isomorphism Theorem, Cauchy's Theorem for Abelian Groups
e. Automorphisms
   Concepts: Automorphism, Inner Automorphism

Chapter 3 – Direct Products and Abelian Groups

a. Examples and Definitions
   Concepts: Direct Product
b. Computing Orders
c. Direct Sums
   Concepts: Direct Sum
d. Fundamental Theorem of Finite Abelian Groups
   Concepts: Decomposable, Square-free
   Principles: Fundamental Theorem of Finite Abelian Groups

Chapter 4 – Group Actions

a. Group Actions and Cayley's Theorem
   Concepts: Group Action,
   Principles: Cayley's Theorem
b. Stabilizers and Orbits under a Group Action
   Concepts: Stabilizer, Orbit
c. Burnside's Theorem and Applications
   Concepts: Fixed
   Principles: Burnside's Theorem

d. Conjugacy Classes and the Class Equation
   Concepts: Conjugate, Conjugacy Class
   Principles: Class Equation
e. Conjugacy in S_n and Simplicity of A_5
   Concepts: Simple

Chapter 14 – Symmetries

a. Linear Transformations
   Concepts: Linear Transformation, Eigenvector, Eigenvalue, Transpose, Dot Product, Length, Distance, Orthogonal Group
b. Isometries
   Concepts: Rigid Motion, Isometry, Translations, Rotations, Reflections
c. Symmetry Groups
   Concepts: Group of Symmetries
   Principles: Fixed Point Theorem
d. Platonic Solids
   Concepts: Regular Polyhedron, Platonic Solid, Dual

Chapter 6 – Rings

a. Examples and Basic Concepts
   Concepts: Ring, Commutative Ring, Ring with Unity, Subring
b. Integral Domains
   Concepts: Zero Divisor, Cancellation Laws, Integral Domain
c. Fields
   Concepts: Unit, Field, Subfield, Division Ring, Characteristic
   Principles: Scarecrow's Theorem

Chapter 7 – Ring Homomorphisms

a. Definitions and Basic Principles
   Concepts: Ring Homomorphism, Kernel, Ring Isomorphism, Isomorphic

b. Ideals
Concepts: Ideal, Principal Ideal
Principles: First Isomorphism Theorem for Rings

Chapter 8 – Ring of Polynomials

a. Basic Concepts and Notation
Concepts: Indeterminate, Polynomial, Value, Argument, Zero, Degree, Leading Coefficient, Rational Function
b. Division Algorithm in $F\,[x]$
Concepts: Quotient, Remainder, Divisor, Multiple, Factoring, Common Divisor, Relatively Prime
Principles: Division Algorithm, Euclidean Algorithm
c. More Applications of the Division Algorithm
Concepts: Multiplicity, nth Roots of Unity
Principles: Factor Theorem, Remainder Theorem
d. Irreducible Polynomials
Concepts: Primitive Polynomial
Principles: Unique Factorization Theorem, Rational Roots Theorem

<u>Grading</u>

Examinations:		
Mid-Term Exams	(September 24, Oct 15, Nov 12)	300 Pts.
Final Exam	(December 15, 1:30-4:30 pm)	200 Pts.
Home Work		
6-10 Assignments, Various Dates		100 Pts.
Writing Assignments		
6-10 Assignments Various Dates, 4 will be returned for rewriting		100 Pts.
Total Grade Point Base		700 Pts.

<u>Scale</u>

A...100% - 90% B...89% - 80% C...79% - 70% D...69% - 60% F...59% - 0%

<u>Critical Dates</u>

September 13	Last Day to Drop on MyTech (Without Penalty)
October 11-12	Fall Break
October 25	Mid-Semester Grades Due
November 1	Last Day to Drop on MyTech (With Penalty)
November 24-25	Thanksgiving Break
December 2-8	Period of No Exams

Notices

Academic Integrity (Extracted from OP 34.12)

It is the aim of the faculty of Texas Tech University to foster a spirit of complete honesty and high standard of integrity. The attempt of students to present as their own any work not honestly performed is regarded by the faculty and administration as a most serious offense and renders the offenders liable to serious consequences, possibly suspension.

"Scholastic dishonesty" includes, but it not limited to, cheating, plagiarism, collusion, falsifying academic records, misrepresenting facts, and any act designed to give unfair academic advantage to the student (such as, but not limited to, submission of essentially the same written assignment for two courses without the prior permission of the instructor) or the attempt to commit such an act.

Observance of Religious Holiday (Extracted from OP 34.19)

A student who intends to observe a religious holy day should make that intention known in writing to the instructor prior to the absence. A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence.

Accommodation for Students with Disabilities

Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Please note instructors are not allowed to provide classroom accommodations to a student until appropriate verification from Student Disability Services has been provided. For additional information, you may contact the Student Disability Services office in 335 West Hall or 806-742-2405.